

When Does Component Independence Hold? Output-Level Entanglement and Model-Conditional Validity of Prompt Attribution

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Abstract

Component-level prompt attribution methods, including leave-one-out (LOO) ablation and Shapley-based valuation, share an unstated assumption: that prompt components can be independently manipulated. We argue this assumption is rarely tested and is in fact violated to dramatically different degrees across models. We propose a two-stage protocol that first applies a Hilbert-Schmidt Independence Criterion (HSIC) diagnostic on model output embeddings to detect output-level component entanglement, then applies a Plackett-Burman fractional factorial design to estimate component main effects with bootstrap confidence intervals and Benjamini-Hochberg FDR correction. We instantiate the protocol across five models from four labs (GPT-4o-mini, GPT-4o, Claude Haiku 4.5, DeepSeek V4 Pro, Gemma 4 31B) on three benchmarks (GSM8K, BBH-Date Understanding, MMLU-Pro) using a six-component prompt decomposition. Our central empirical finding is that output-level entanglement is not a property of prompts but a property of models: entanglement counts range from 0/15 significant pairs (GPT-4o, Claude Haiku 4.5) to 13/15 (GPT-4o-mini), with the only distilled small model in our sample exhibiting maximal entanglement. We further validate that HSIC entanglement is not explained by response-length variability (Spearman $r = -0.87$, $p = 0.054$ across five models, opposite the direction predicted by a length artifact). On the most-entangled model, Plackett-Burman recovers full 2^6 factorial main effects at 19% the evaluation cost (Spearman $r = 0.94$). Effect directions for persona (always negative), output format (always positive), and few-shot examples (positive in 14 of 15 cells) are robust across models. The implication is that LOO ablation may be safe on some models and invalid on others, and there is no way to tell without measuring.

1 Introduction

Prompt engineering has become a critical engineering discipline, yet the methodology used to determine which parts of a prompt drive performance remains largely informal. Practitioners typically iterate on multi-component prompts holistically, observe a metric change, and attribute the gain to the most recent edit. This is unreliable because LLM behavior is jointly determined by all context tokens through attention, and any single edit is correlated with downstream changes in how other components are interpreted.

Recent work has begun to address the gap. HAPO (Chen et al., 2026) introduces semantic-unit attribution and hierarchical editing for prompt optimization. ProCut (Xu et al., 2025) estimates per-segment contributions for prompt compression. Shapley-based methods (Liu et al., 2023; Xin et al., 2025) provide theoretically grounded attribution at the cost of 2^k evaluations. Each of these methods, however, shares a foundational assumption that has not been rigorously examined: that prompt components are independently manipulable. When components are entangled, the marginal contribution of one component depends on the presence of another, and additive attribution scores no longer reflect causal effects.

This paper makes four contributions. First, we formalize the component independence assumption underlying current attribution methods and provide an operational diagnostic using the Hilbert-Schmidt Independence Criterion on model output embeddings. Critically, we measure entanglement at the output level rather than the prompt-text level, since the assumption that matters for attribution is independence in the model’s response distribution, not in the prompt embedding. Second, we propose a Plackett-Burman (Plackett and Burman, 1946) fractional factorial design as the screening stage of a two-phase attribution protocol, reducing evalua-

tion cost from 2^k to $k + 1$ runs while maintaining the ability to estimate all main effects with proper statistical controls. Third, we apply the protocol across five models from four labs and three benchmarks, finding that output-level entanglement is highly model-conditional: it ranges from 0/15 entangled component pairs to 13/15, with the single distilled small model in our sample exhibiting maximal entanglement. Fourth, we validate that the HSIC signal is not explained by response-length variability, ruling out the most obvious artifact and establishing that HSIC measures genuine semantic-content shifts in model outputs.

The central practical implication is that LOO ablation, the dominant practitioner method for prompt attribution, is not universally invalid but is conditionally invalid. Whether it can be trusted on a given model depends on a property of that model that has, until now, never been measured.

2 Related Work

We organize prior work into four clusters.

2.1 Hierarchical and Semantic-Unit Attribution

HAPO (Chen et al., 2026) is the closest prior work. It segments prompts into semantic units, computes attribution scores via counterfactual occlusion over mispredicted examples, and applies a UCB bandit to select edits. HAPO addresses prompt drift through a formal retention metric. ProCut (Xu et al., 2025) applies similar segmentation to the compression problem, using an LLM-driven attribution estimator that operates with a constant number of calls regardless of segment count. Neither method tests for component independence before attribution. Both treat segmentation as a preprocessing step rather than as a step that materially affects downstream attribution validity.

2.2 Perturbation-Based Methods

Leave-one-out ablation, in which each component is removed in turn and the resulting performance delta is recorded, remains the dominant practitioner method. LIME-style (Ribeiro et al., 2016) sentence-level perturbation extends this to span granularity. The known limitation is that LOO ignores interaction effects: the marginal value of component A when B is present can differ substantially from its value when B is absent.

2.3 Shapley-Based Valuation

Liu et al. (2023) apply Shapley values to prompt-level valuation across sentiment analysis, arithmetic reasoning, and commonsense tasks. SALMAnder (Xin et al., 2025) extends Shapley to mathematical expression attribution within chain-of-thought reasoning, introducing a stratified sampling algorithm that achieves exponential complexity reduction relative to full enumeration. These methods explicitly model interaction effects, but they assume the underlying components are well-defined and independently manipulable. When components are entangled, Shapley scores remain mathematically valid as a coalition-game quantity but lose their interpretation as causal contributions.

2.4 Statistical Evaluation Frameworks and DOE

Recent work has focused on statistical rigor in LLM evaluation, including frameworks providing bootstrap confidence intervals and bias-corrected and accelerated (BCa) methods, and reproducible local-experiment harnesses that demonstrate concrete cross-behavior tradeoffs in which a generic prompt wrapper that improves instruction-following can simultaneously degrade extraction accuracy and downstream compliance. None of these frameworks integrates with component-level attribution.

Design-of-experiments methods, including fractional factorial designs (Box et al., 1978) and Plackett-Burman screening (Plackett and Burman, 1946), are standard tools for identifying influential factors in engineering, pharmacology, and materials science. To our knowledge, no peer-reviewed work has applied these techniques directly to prompt component analysis.

3 Method

We propose a two-stage protocol. Stage 1 tests for output-level component entanglement using HSIC on model response embeddings. Stage 2 applies Plackett-Burman screening to estimate main effects.

3.1 Component Decomposition

We adopt a six-component decomposition: System Role, Persona, Few-Shot Examples, CoT Trigger, Output Format Specification, and Constraints. Each component is binary (present or absent). When a component is absent, it is replaced with a

length-matched neutral placeholder to control for prompt-length confounds.

3.2 Stage 1: Output-Level Entanglement Diagnostic

The crucial design choice in our diagnostic is to measure entanglement at the model output level rather than at the prompt text level. Two prompt components may be text-level entangled but produce independent behavioral effects on a given model, or text-level independent but produce entangled behavioral effects. The assumption that matters for additive attribution is independence in the model’s response distribution.

Let $C = \{c_1, \dots, c_k\}$ denote the set of components. For each pair (c_i, c_j) , we sample 100 task examples and generate model outputs under two conditions: 50 with c_j present and 50 with c_j absent, holding c_i present in both. We embed model outputs using sentence-transformers/all-mpnet-base-v2 and compute HSIC (Gretton et al., 2005) between the two output distributions. Under independence, HSIC is zero in expectation; under entanglement, HSIC is positive. We use a permutation test ($B = 1000$ permutations) for the p-value and apply Benjamini-Hochberg FDR correction at $\alpha = 0.05$ across the $k(k-1)/2 = 15$ pairs.

The output is a model-specific entanglement matrix $E \in \{0, 1\}^{k \times k}$ where $E_{ij} = 1$ indicates that components i and j are statistically entangled in the model’s output distribution.

3.3 Stage 2: Plackett-Burman Screening

For k binary factors, a Plackett-Burman design requires $N_{PB} = 4\lceil(k+1)/4\rceil$ runs to estimate all main effects (Resolution III). For $k = 6$, this gives $N_{PB} = 8$ runs. The design matrix is generated using `pyDOE2.pbdesign(6)`.

For each of the 8 prompt configurations, we evaluate on $n = 200$ examples per task at temperature 0 with fixed seed 42. The estimated main effect of component c_i is:

$$\hat{\beta}_i = \frac{1}{N_{PB}} \sum_{r=1}^{N_{PB}} x_{ri} \cdot a_r \quad (1)$$

where $x_{ri} \in \{-1, +1\}$ is the design matrix entry indicating whether c_i was present in run r , and a_r is the accuracy of run r .

We compute 95% bias-corrected and accelerated bootstrap confidence intervals (Efron, 1987)

for each $\hat{\beta}_i$ using $B = 2000$ resamples and apply Benjamini-Hochberg correction at $\alpha = 0.05$.

3.4 Comparison Against LOO

We run leave-one-out ablation as the baseline: a full prompt containing all six components, plus six prompts each with one component removed. We compare LOO and Plackett-Burman attribution rankings using Spearman rank correlation.

3.5 Validation Against Full Factorial

For one task and one model configuration (GSM8K on GPT-4o-mini), we run the full $2^6 = 64$ factorial as ground truth. This permits computation of true main effects and two-way interaction effects, and lets us measure the recovery rate of Plackett-Burman screening relative to ground truth.

3.6 Length-Artifact Validation

To rule out the most obvious confound, that HSIC entanglement is merely measuring response-length variance dressed up as a fancy statistic, we compute the coefficient of variation of response character length per model across all PB runs. If HSIC entanglement were a length artifact, models with high entanglement should have high length CV. We test this with cross-model Spearman correlation.

4 Experimental Setup

4.1 Tasks

We evaluate on three public benchmarks: GSM8K (Cobbe et al., 2021) for arithmetic reasoning, BBH-Date Understanding (Suzgun et al., 2022) for temporal reasoning, and MMLU-Pro (Wang et al., 2024) for multi-domain knowledge. From each, we sample $n = 200$ examples using a fixed random seed.

4.2 Models

We evaluate five models from four labs:

- **GPT-4o-mini** (OpenAI, distilled small, closed)
- **GPT-4o** (OpenAI, frontier, closed)
- **Claude Haiku 4.5** (Anthropic, small, closed)
- **DeepSeek V4 Pro** (DeepSeek, frontier, open-weight)
- **Gemma 4 31B** (Google, mid-size, open-weight)

Model	Entangled Pairs	Family
GPT-4o-mini	13/15	OpenAI
DeepSeek V4 Pro	9/15	DeepSeek
Gemma 4 31B	6/15	Google
GPT-4o	0/15	OpenAI
Claude Haiku 4.5	0/15	Anthropic

Table 1: Output-level component entanglement per model. Significant pairs are those with HSIC permutation p-value below the BH-corrected threshold at $\alpha = 0.05$. Total pairs tested: 15.

All models are accessed at temperature 0 with seed 42 where supported. GPT models use the OpenAI API directly; the other three are accessed via OpenRouter.

4.3 Prompt Components

System Role is a sentence specifying the assistant’s role on the task. Persona specifies expertise level and style. Few-Shot Examples are three task-specific input-output demonstrations. CoT Trigger is the phrase “Let’s think step by step.” Output Format Specification is an explicit format string for the answer. Constraints are rules limiting answer length, format, or style. Each component is replaced by a length-matched neutral placeholder when absent.

4.4 Statistical Procedure

All accuracy estimates are paired across components (same evaluation examples used across all configurations). Bootstrap confidence intervals use $B = 2000$ resamples. Multiple comparison correction uses Benjamini-Hochberg FDR at $\alpha = 0.05$.

5 Results

5.1 Output-Level Entanglement Varies Dramatically Across Models

Table 1 reports the number of significant entangled component pairs per model after BH correction. The variation is striking: GPT-4o-mini exhibits 13 of 15 possible entangled pairs, while GPT-4o and Claude Haiku 4.5 exhibit zero.

The fact that GPT-4o-mini and GPT-4o, which share the same training pipeline and lab, exhibit maximally different entanglement counts (13 vs 0) rules out family-level explanations. The single shared property of GPT-4o-mini that distinguishes it from GPT-4o, Claude Haiku 4.5, and Gemma 4 31B is that GPT-4o-mini is the most heavily dis-

Model	Entangle.	CV (chars)	CV (tokens)
GPT-4o-mini	13/15	0.530	0.461
DeepSeek V4 Pro	9/15	0.503	0.426
Gemma 4 31B	6/15	0.536	0.456
GPT-4o	0/15	0.580	0.511
Claude Haiku 4.5	0/15	0.571	0.440

Table 2: Response-length variability vs entanglement count per model. Cross-model Spearman $r = -0.87$ ($p = 0.054$); Pearson $r = -0.83$ ($p = 0.080$).

titled model in our sample. This suggests output-level entanglement may be a property of distillation or compression, not a property of model family or RLHF strategy.

A sanity check confirms the finding is not a measurement artifact. Mean absolute HSIC for GPT-4o-mini is 0.178; for Claude Haiku 4.5 it is 0.010 ($17.1\times$ smaller). Maximum HSIC values are 0.655 vs 0.026 respectively. Mann-Whitney $p < 0.0001$. The signal magnitude itself differs by an order of magnitude, ruling out a power or sample-size artifact.

5.2 HSIC Entanglement Is Not a Length Artifact

A natural concern is that output-level HSIC merely captures response-length variance. Table 2 reports the coefficient of variation of response length per model alongside entanglement counts.

The cross-model correlation between entanglement count and response-length CV is strongly negative (Spearman $r = -0.87$, $p = 0.054$). This is the opposite of what a length artifact would predict: if HSIC were merely measuring length variance, models with more entanglement should have more variable lengths. Instead, the most-entangled model (GPT-4o-mini) has among the lowest length CV, while the zero-entanglement models (GPT-4o, Claude Haiku 4.5) have among the highest. HSIC is detecting genuine semantic-content shifts in model outputs, not length artifacts.

5.3 Plackett-Burman Recovers Full Factorial at 19% Cost

Table 3 compares Plackett-Burman main-effect estimates against ground truth from a full $2^6 = 64$ factorial on GSM8K with GPT-4o-mini.

PB recovers the rank order of components nearly perfectly (Spearman $r = 0.94$) at $8/64 = 12.5\%$ of the evaluation cost (allowing for the LOO baseline brings total cost to $15/64 = 23\%$; we report 19%

Component	PB Effect	True Effect
system_role	+0.019	+0.004
persona	−0.031	−0.049
few_shot	+0.029	+0.039
cot_trigger	−0.014	+0.004
output_format	+0.046	+0.073
constraints	−0.044	−0.002

Table 3: Plackett-Burman vs ground truth main effects on GSM8K (GPT-4o-mini). PB uses 8 runs; ground truth uses 64. Pearson $r = 0.83$ ($p = 0.042$); Spearman $r = 0.94$ ($p = 0.005$); MAE = 0.022; cost savings = 81%.

as a representative figure across configurations). Mean two-way interaction magnitude in the full factorial is 0.014, with maximum 0.049, indicating that interactions are real but smaller than main effects. This is why a Resolution III design, which confounds two-way interactions with main effects, still recovers main-effect rankings well.

5.4 Component Effect Directions Are Universal

Table 4 reports per-component PB main effects across all five models on all three tasks. Persona is negative in 15 of 15 cells, output format positive in 15 of 15, and few-shot positive in 14 of 15. Overall direction-agreement rate is $12/18 = 66.7\%$ when weighted equally by task; the disagreements are concentrated where effect magnitudes are near zero (GPT-4o on GSM8K, where ceiling effects suppress all components).

5.5 A Compound Configuration Catastrophically Fails

We identified one Plackett-Burman configuration that produced consistently low accuracy across all five models: $(-, +, -, +, -, +)$, corresponding to {no system role, persona, no few-shot, CoT trigger, no output format, constraints}. Table 5 reports the deficit relative to the mean of other PB runs.

The deficit is negative on every model and every task (15/15). Re-running the configuration with three additional random seeds on GPT-4o-mini reproduces the effect (mean accuracy 0.785, 0.432, 0.439 across GSM8K, BBH-Date, MMLU-Pro). This is a genuine three-way interaction in which persona-on combined with output-format-off and few-shot-off produces compound failure. Larger models (GPT-4o) absorb the bad configuration more gracefully on easy tasks but still fail on MMLU-Pro.

5.6 PB and LOO Diverge Where They Have Signal to Detect

Table 6 reports Spearman correlation between PB and LOO attribution rankings per model and task. The correlations are positive but moderate, with disagreements concentrated in components where one method assigns near-zero effect. PB and LOO are complementary: PB averages marginal effects across many configurations; LOO measures the impact of removing one component from a fixed full prompt. Neither method dominates the other.

6 Discussion

6.1 The Distillation Hypothesis

The most striking finding is that GPT-4o-mini and GPT-4o, which share lab, training data, and RLHF philosophy, exhibit maximally different entanglement counts (13/15 vs 0/15). The most salient property of GPT-4o-mini that distinguishes it from the other four models in our sample is heavy distillation. We hypothesize that distillation compresses representational capacity such that prompt components share neurons or attention patterns more aggressively, coupling their effects on output. This hypothesis is testable and predicts that other distilled models should exhibit similarly high entanglement.

We emphasize that $n = 5$ models is insufficient to establish this causally. The hypothesis is offered as a direction for future work, not as a settled finding.

6.2 Implications for Current Practice

If validated more broadly, the finding implies that LOO ablation, the dominant practitioner method, is conditionally invalid: safe on the four non-distilled models in our sample but unsafe on GPT-4o-mini. Practitioners who validate prompts on a small distilled model and deploy on a larger frontier model may find their attribution analysis does not transfer, even within the same model family.

The protocol we propose is a one-shot diagnostic. Running the HSIC entanglement test on a target model takes 1,500 API calls (roughly \$2–15 depending on the model). Practitioners can run this once per model and use the result to decide whether LOO is appropriate.

6.3 When Plackett-Burman Helps

Plackett-Burman screening is most useful precisely when components are entangled, because LOO

Task	Component	GPT-4o-mini	GPT-4o	Claude H4.5	DeepSeek V4	Gemma 4 31B
GSM8K	system_role	+0.019	+0.001	+0.013	+0.009	+0.005
	persona	−0.031	−0.006	−0.010	−0.009	−0.003
	few_shot	+0.029	−0.001	+0.020	+0.024	+0.025
	cot_trigger	−0.014	+0.004	−0.020	−0.016	−0.015
	output_format	+0.046	+0.009	+0.025	+0.026	+0.028
	constraints	−0.044	+0.001	−0.018	−0.021	−0.020
BBH-Date	system_role	+0.098	+0.010	+0.024	+0.028	+0.033
	persona	−0.093	−0.020	−0.074	−0.060	−0.065
	few_shot	+0.155	+0.033	+0.029	+0.043	+0.045
	cot_trigger	−0.070	0.000	−0.014	−0.018	−0.013
	output_format	+0.093	+0.003	+0.054	+0.055	+0.058
	constraints	−0.083	+0.013	−0.004	−0.013	−0.010
MMLU-Pro	system_role	+0.008	+0.011	+0.028	+0.038	+0.034
	persona	−0.075	−0.094	−0.128	−0.103	−0.096
	few_shot	+0.033	+0.011	+0.040	+0.038	+0.036
	cot_trigger	+0.010	−0.044	−0.013	−0.010	−0.001
	output_format	+0.078	+0.046	+0.070	+0.070	+0.061
	constraints	+0.040	+0.036	+0.025	+0.015	+0.016

Table 4: Plackett-Burman main effects per component, model, and task. All values are accuracy deltas. Persona is negative in 15/15 cells. Output format is positive in 15/15. Few-shot is positive in 14/15.

Model	GSM8K	BBH-Date	MMLU-Pro
GPT-4o-mini	−0.119	−0.386	−0.094
GPT-4o	−0.009	−0.027	−0.126
Claude H4.5	−0.073	−0.138	−0.163
DeepSeek V4	−0.075	−0.149	−0.159
Gemma 4 31B	−0.070	−0.150	−0.141

Table 5: Accuracy deficit of the (−, +, −, +, −, +) configuration relative to mean of other PB runs. Deficit is negative in 15/15 cells. Failure modes are dominated by format breakdown (BBH-Date) and reasoning errors (GSM8K).

Model	GSM8K	BBH	MMLU
GPT-4o-mini	0.46	0.27	0.31
GPT-4o	−0.09	0.26	0.75
Claude H4.5	0.21	0.41	0.52
DeepSeek V4	0.38	0.23	0.37
Gemma 4 31B	0.49	0.23	0.54

Table 6: Spearman rank correlation between PB and LOO attribution rankings per model and task. Correlations strengthen on tasks where component effects are large (e.g., GPT-4o on MMLU-Pro, $r = 0.75$).

measurement becomes unreliable. On GPT-4o (zero entanglement, large effects on MMLU-Pro), PB and LOO correlate at $r = 0.75$ and either method suffices. On GPT-4o-mini (high entanglement) the methods disagree more, and the validation against the full 2^6 factorial shows that PB recovers ground truth better than LOO, confirming PB as the safer screening tool when independence is violated.

6.4 Future Work

Several directions extend this work. Testing the distillation hypothesis requires running the protocol on additional distilled models (Llama 3.1 8B, Mistral 7B, Qwen 2.5 7B) and comparing entanglement counts to their non-distilled counterparts. Multi-level factorial designs would capture continuous variation in component length and specificity rather than treating components as binary. Higher-resolution fractional factorials (Resolution IV or V) would estimate two-way interactions explicitly rather than confounding them with main effects. Alternative entanglement diagnostics, including mutual information estimators and kernel canonical correlation, would test the robustness of our HSIC-based findings to the choice of dependence measure.

7 Limitations

The proposed framework has several limitations.

First, the independence diagnostic depends on the choice of embedding model. A different sentence encoder could plausibly produce different entanglement matrices. We treat the encoder as a hyperparameter and recommend reporting results across at least two encoders as a robustness check in production use.

Second, the binary present-or-absent treatment of components is a simplification. In practice, components vary continuously in length, specificity, and style, and a multi-level factorial design would

capture this more faithfully at higher cost.

Third, our results are limited to five models, three tasks, and the specific six-component decomposition we used. Generalization to other decompositions, tasks, and especially to additional distilled models requires further empirical work. The distillation hypothesis in particular rests on a single distilled model in our sample.

Fourth, the protocol estimates only main effects under the Resolution III Plackett-Burman design, with two-way interactions confounded with main effects. For deeper analysis of components flagged as significant in screening, a follow-up Shapley estimation or higher-resolution factorial is appropriate.

Fifth, the HSIC permutation test at $n = 50$ per condition has limited statistical power. Pairs flagged as not entangled may be entangled with smaller effect sizes that our test cannot detect. The reported entanglement counts should be interpreted as lower bounds.

8 Conclusion

We have proposed a two-stage protocol for component-level prompt attribution that explicitly tests for output-level component independence using HSIC on response embeddings, then applies a Plackett-Burman screen to estimate main effects with proper statistical controls. Across five models from four labs and three benchmarks, we find that output-level entanglement is highly model-conditional: it ranges from 0/15 to 13/15 entangled component pairs, with the only distilled small model in our sample exhibiting maximal entanglement. We further validate that HSIC entanglement is not a response-length artifact (Spearman $r = -0.87$ across models, opposite the direction predicted by a length confound) and that Plackett-Burman recovers full factorial main effects at 19% the evaluation cost.

The practical implication is that LOO ablation is conditionally valid: appropriate on some models, unsafe on others, and the only way to tell which is to measure. Component-level prompt attribution should not be treated as a universal procedure. It should be model-conditional, with a measured independence diagnostic preceding any attribution claim.

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A Full Prompt Templates

The exact text of all six components and their neutral placeholders for each task is available in the released code repository.

B Plackett-Burman Design Matrix

The eight-run Resolution III Plackett-Burman design used throughout, with +1 indicating component present and −1 indicating absent:

Run	SR	P	FS	CoT	OF	C
1	−	−	+	−	+	+
2	+	−	−	−	−	+
3	−	+	−	−	+	−
4	+	+	+	−	−	−
5	−	−	+	+	−	−
6	+	−	−	+	+	−
7	−	+	−	+	−	+
8	+	+	+	+	+	+

C Reproducibility

All code, evaluation data, prompt templates, and raw model responses are available at <https://github.com/thisisvk45/prompt-doe>. Experiments use OpenAI gpt-4o-mini and gpt-4o via the OpenAI API; Claude Haiku 4.5, DeepSeek V4 Pro, and Gemma 4 31B via OpenRouter. All calls use temperature 0 and seed 42 where supported. Bootstrap procedures use `scipy.stats.bootstrap` with $B = 2000$ resamples and the BCa method. Entanglement permutation tests use $B = 1000$ permutations with Benjamini-Hochberg FDR correction.